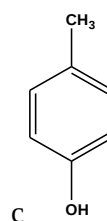
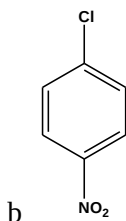
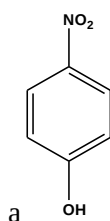
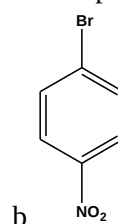
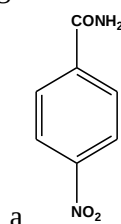


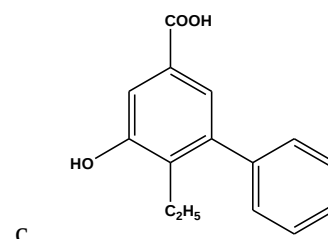
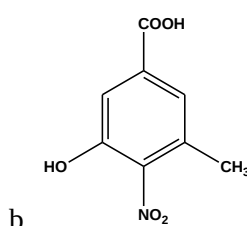
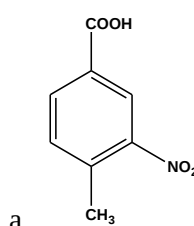
- Calculate the logP value for the following compounds if (logP value of benzene = 2.13, compound a $\Pi_{\text{NO}_2} = 0.5$ and $\Pi_{\text{OH}} = 0.11$; compound b $\Pi_{\text{Cl}} = 0.54$ and $\Pi_{\text{NO}_2} = 0.25$ compound c $\Pi_{\text{CH}_3} = 0.49$ and $\Pi_{\text{OH}} = -0.15$)



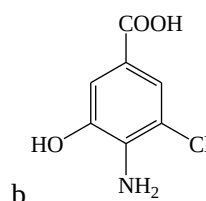
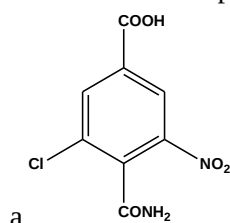
- Calculate the lipophilic substituent constant of benzamide and bromine in the following compounds respectively if (logP value of benzene = 2.13, logP value of p-Nitrobenzamide = 1.65, logP value of p-bromo nitrobenzene = 2.43 and Π_{NO_2} in compounds a = 0.46, b = -0.32)



- Calculate the Pk of the following compounds if the hammetts constants (σ) of $\text{CH}_3(\text{m}) = -0.12$ and $\text{CH}_3(\text{p}) = -0.17$, $\text{NO}_2(\text{m}) = 0.71$ and $\text{NO}_2(\text{p}) = 0.78$, $\text{OH}(\text{m}) = 0.12$, $\text{C}_2\text{H}_5(\text{p}) = -0.15$, $\text{C}_6\text{H}_5(\text{m}) = 0.16$ (P^{KH} of benzoic acid = 4.2)



- Calculate the Hammett constants (σ) of CONH_2 and NH_2 in the following compounds respectively if (P^{KH} of benzoic acid = 4.2, $\text{Cl}(\text{m}) = 0.37$, $\text{NO}_2(\text{m}) = 0.71$, $\text{OH}(\text{m}) = 0.12$; Pk of compound a & b are 3.65 and 3.95 respectively)



- Calculate the rate constant of the following compounds if (rate constant of methylethanoate = 1.26)
 - $\text{CH}_3\text{OCH}_2\text{COOCH}_3$
 - $\text{PhOCH}_2\text{CH}_2\text{COOCH}_3$
 - $\text{CH}_3\text{SCH}_2\text{COOCH}_3$
 - $\text{F}_3\text{CCH}_2\text{COOCH}_3$
- Calculate the biological activity and the minimum concentration required to cause a defined biological response of the drug with a wide range of P values as $P = 0.978$ if (constants $k_1=0.60$, $k_2=1.23$ and $k_3=1.54$)
- Calculate the biological activity and the minimum concentration required to cause a defined biological response of the drug with $P=0.965$, $\sigma=-0.58$ & $E_s=0.77$; with numerical constants $k_1=1.04$, $k_2=0.99$ and $k_3=1.45$ and $k_4=0.85$
- Does Lipinski's "rule of five" predict good oral bioavailability for the following drugs? (Evaluate them by Lipinski's "rule of five").
 - Ketoconazol (Mlogpvalue = 4.45)
 - Chloramphenicol (Mlogpvalue = 1.23)
 - Quinine (Mlogpvalue = 2.8)
 - Itraconazole (Mlog pvalue = 5.5)